MD. SHAHINUR HASAN

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Career Objective

Aspiring AI/ML Engineer and ECE graduate with strong interest in multimodal AI systems, computer vision, NLP, and intelligent compliance automation. Experienced in machine learning, LLM-based systems, vector search, and data engineering workflows. Seeking to contribute technical and analytical skills to AI-driven electrical safety and standards-based compliance solutions.

Professional Competency

Programming Languages:

Python, C, C++, SQL.

Machine Learning & AI:

Machine Learning, Deep Learning, Model Evaluation, Feature Engineering, Supervised Learning, Unsupervised Learning, Image Processing.

Generative AI & LLMs:

LLMs, Prompt Engineering, RAG (Retrieval-Augmented Generation), Embeddings, Vector Search, Fine-Tuning.

Frameworks & Libraries , Databases:

Scikit-learn, Pandas, NumPy, Matplotlib, TensorFlow, PyTorch (basic), LangChain.

Databases :

MySQL, NoSQL(basic).

Education

Degree	CGPA/GPA	Institution	Year
B.Sc. in Electronics and Communication Engineering (ECE)	3.64 / 4.00	Khulna University of Engineering & Technology (KUET)	2026
HSC	5.00 / 5.00	Netrakona Govt. College	2020
SSC	5.00 / 5.00	Baniajan C.T. Pilot High School	2018

Certifications

1. **Introduction to Python** — **DataCamp**
   Learned Python fundamentals including data types, control flow, functions, loops, and basic data manipulation for programming and data analysis.
2. **Supervised Machine Learning** — **Coursera**
   Completed Regression and Classification Algorithms, Cost Functions, Model Evaluation, and Gradient Descent using NumPy and Scikit-learn.
3. **Advanced Learning Algorithm** — **Coursera**
   Completed coursework on Neural Networks, TensorFlow, Decision Trees, Ensemble Methods, machine learning lifecycle, and model optimization.
4. **Neural Networks and Deep Learning** — **Coursera**
   Completed coursework on Forward & Backpropagation, CNN fundamentals and Hyperparameter Tuning , Transfer Learning using TensorFlow.
5. **Sequence Models** — **Coursera – Continued**
   Learning RNNs, LSTMs, GRUs, Attention Mechanisms, NLP pipelines, and sequence modeling techniques using TensorFlow and Python.

Projects

1. **Multimodal Building Crack Detection** — **GitHub**
   Features: Developed a multimodal deep learning system for structural crack detection in concrete, fusing visual features from a fine-tuned ResNet50 with acoustic features extracted via mel-spectrogram CNN using a gated attention fusion mechanism.
2. **Threat-Lens** — **GitHub**
   Features: Designed a scalable ETL pipeline for a security system, integrating data ingestion, transformation, validation with MongoDB database and artifact layers. Developed Python workflows integrating MLflow and DagsHub for experiment tracking, deployed via Docker on AWS EC2/S3.
3. **Intent Based Searching** — **GitHub**
   Features:A hybrid semantic search system using Sentence-Transformer, FAISS, and BM25 for improved retrieval accuracy. Developed embedding pipelines with Redis caching for low-latency inference; contributed to data preprocessing and transformation in a hackathon arranged by Poridhi.io.
4. **Multi-Document RAG Assistant** — **GitHub**
   Features: Built a scalable RAG-based QA system leveraging transformer embeddings, FAISS vector search, and LangChain orchestration, enabling low-latency semantic querying over multiple PDF documents via an interactive Streamlit interface.

Publications

1. Graph-Based Approaches to Social Network Analysis: Pinpointing Central Nodes. (QPIIN-2026)
2. CHRONICLE: Continual Heterogeneous Graph Neural Learning for Incremental Cyberattack Classification on Blockchain. (Under review)

Achievements

3. Dean's award (2023)
4. 16th in DU AI CHALLENGE arranged by RMEDU Student Club, Dhaka University.

Extra-Curricular Activities

1. Joined several AI hackathon and Datathon arranged by different clubs in BD.
2. KUET Mars Rover.
   - Member, Software Sub-Team.